pyNoxtli

Luis M. de la Cruz

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Índice

1. Modelo numérico 1D **2**
1.1. Ecuaciones FVM 2
1.2. Almacenamiento de la matriz: CSR 3
1.3. Condiciones de frontera 3
1.3.1. Dirichlet 4
1.3.2. Neumman 4
1.3.3. Ajuste en el almacenamiento CSR 4

2. Modelo numérico 2D **5**
2.1. Ecuaciones FVM 5
2.2. Almacenamiento de la matrix: CSR 7
2.3. Condiciones de frontera 9
2.3.1. Dirichlet 9
2.3.2. Neumman 10
2.3.3. Doble condición de frontera 12
2.3.4. Ajuste en el almacenamiento CSR 13

3. Numerical Schemes **14**

1. Modelo numérico 1D

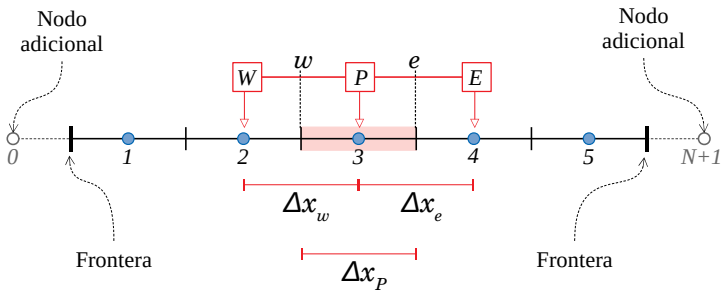

Figura 1: Malla en 1D.

$$\boldsymbol{a_P}\psi_P = \boldsymbol{a_E}\psi_E + \boldsymbol{a_W}\psi_W + \boldsymbol{b_P} \quad \Longrightarrow \quad -\boldsymbol{a_W}\psi_W + \boldsymbol{a_P}\psi_P - \boldsymbol{a_E}\psi_E = \boldsymbol{b_P}$$

Para evitar confusión en los coeficientes $\boldsymbol{a}$'s, usarémos un superíndice indicando a qué volumen corresponde la ecuación. Por ejemplo para $\boldsymbol{P} = 3$ se tiene:

$$-\boldsymbol{a}^3_{\boldsymbol{W}}\psi_2 + \boldsymbol{a}^3_{\boldsymbol{P}}\psi_3 - \boldsymbol{a}^3_{\boldsymbol{E}}\psi_4 = \boldsymbol{b}_3$$

1.1. Ecuaciones FVM

$$\begin{aligned}
\boldsymbol{W} = 0, \boldsymbol{P} = 1, \boldsymbol{E} = 2 \quad &: \quad -\boldsymbol{a}^1_{\boldsymbol{W}}\psi_0 + \boldsymbol{a}^1_{\boldsymbol{P}}\psi_1 - \boldsymbol{a}^1_{\boldsymbol{E}}\psi_2 = \boldsymbol{b}_1 \\
\boldsymbol{W} = 1, \boldsymbol{P} = 2, \boldsymbol{E} = 3 \quad &: \quad -\boldsymbol{a}^2_{\boldsymbol{W}}\psi_1 + \boldsymbol{a}^2_{\boldsymbol{P}}\psi_2 - \boldsymbol{a}^2_{\boldsymbol{E}}\psi_3 = \boldsymbol{b}_2 \\
\boldsymbol{W} = 2, \boldsymbol{P} = 3, \boldsymbol{E} = 4 \quad &: \quad -\boldsymbol{a}^3_{\boldsymbol{W}}\psi_2 + \boldsymbol{a}^3_{\boldsymbol{P}}\psi_3 - \boldsymbol{a}^3_{\boldsymbol{E}}\psi_4 = \boldsymbol{b}_3 \\
\boldsymbol{W} = 3, \boldsymbol{P} = 4, \boldsymbol{E} = 5 \quad &: \quad -\boldsymbol{a}^4_{\boldsymbol{W}}\psi_3 + \boldsymbol{a}^4_{\boldsymbol{P}}\psi_4 - \boldsymbol{a}^4_{\boldsymbol{E}}\psi_5 = \boldsymbol{b}_4 \\
\boldsymbol{W} = 4, \boldsymbol{P} = 5, \boldsymbol{E} = 6 \quad &: \quad -\boldsymbol{a}^5_{\boldsymbol{W}}\psi_4 + \boldsymbol{a}^5_{\boldsymbol{P}}\psi_5 - \boldsymbol{a}^5_{\boldsymbol{E}}\psi_6 = \boldsymbol{b}_5
\end{aligned}$$

$$\left(\begin{array}{c|ccccc|c}
\boxed{-\boldsymbol{a}^1_{\boldsymbol{W}}} & \boldsymbol{a}^1_{\boldsymbol{P}} & -\boldsymbol{a}^1_{\boldsymbol{E}} & 0 & 0 & 0 & 0 \\
0 & -\boldsymbol{a}^2_{\boldsymbol{W}} & \boldsymbol{a}^2_{\boldsymbol{P}} & -\boldsymbol{a}^2_{\boldsymbol{E}} & 0 & 0 & 0 \\
0 & 0 & -\boldsymbol{a}^3_{\boldsymbol{W}} & \boldsymbol{a}^3_{\boldsymbol{P}} & -\boldsymbol{a}^3_{\boldsymbol{E}} & 0 & 0 \\
0 & 0 & 0 & -\boldsymbol{a}^4_{\boldsymbol{W}} & \boldsymbol{a}^4_{\boldsymbol{P}} & -\boldsymbol{a}^4_{\boldsymbol{E}} & 0 \\
0 & 0 & 0 & 0 & -\boldsymbol{a}^5_{\boldsymbol{W}} & \boldsymbol{a}^5_{\boldsymbol{P}} & \boxed{-\boldsymbol{a}^5_{\boldsymbol{E}}}
\end{array}\right)
\begin{pmatrix} \psi_0 \\ \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \\ \psi_5 \\ \psi_6 \end{pmatrix}
=
\begin{pmatrix} \boldsymbol{b}_1 \\ \boldsymbol{b}_2 \\ \boldsymbol{b}_3 \\ \boldsymbol{b}_4 \\ \boldsymbol{b}_5 \end{pmatrix}$$

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1.2. Almacenamiento de la matriz: CSR

Algoritmo 1 Formato CSR IJK 1D

```
INPUT: bi, ei
OUTPUT: AA, JA, IA
I := 0
irow := 0
IA[irow] := I
for i := bi to ei do
    Calculate a^i_W, a^i_P, a^i_E
    AA[I] := a^i_W
    JA[I] := irow - 1
    I := I + 1
    AA[I] := a^i_P
    JA[I] := irow
    I := I + 1
    AA[I] := a^i_E
    JA[I] := irow + 1
    I := I+ 1
    irow := irow + 1
    IA[irow] := I
end for
JA[0] := 0
JA[Nx-1] := 0
```

Matriz del sistema.

i	0	1	2	3	4
0	a^1_P	$-a^1_E$	0	0	0
1	$-a^2_W$	a^2_P	$-a^2_E$	0	0
2	0	$-a^3_W$	a^3_P	$-a^3_E$	0
3	0	0	$-a^4_W$	a^4_P	$-a^4_E$
4	0	0	0	$-a^5_W$	a^5_P

Resultado del almacenamiento:

AA	=	$\boxed{-a^1_W}$	a^1_P	$-a^1_E$	$-a^2_W$	a^2_P	$-a^2_E$	$-a^3_W$	a^3_P	$-a^3_E$	$-a^4_W$	a^4_P	$-a^4_E$	$-a^5_W$	a^5_P	$\boxed{-a^5_E}$
JA	=	$\boxed{0}$	0	1	0	1	2	1	2	3	2	3	4	3	4	$\boxed{0}$
IA	=	0	3	6	9	12	15									

1.3. Condiciones de frontera

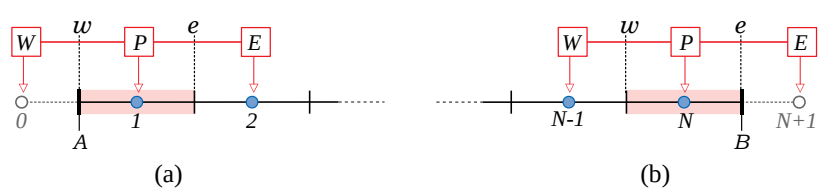

Figura 2: Malla en 1D.

1.3.1. Dirichlet

$$
\begin{aligned}
\frac{\psi_0+\psi_1}{2} &= \psi_A \\
\psi_0 &= 2\psi_A-\psi_1 \\
-a_W^1\psi_0+a_P^1\psi_1-a_E^1\psi_2 &= b_1 \\
-a_W^1(2\psi_A-\psi_1)+a_P^1\psi_1-a_E^1\psi_2 &= b_1 \\
(a_P^1+a_W^1)\psi_1-a_E^1\psi_2 &= b_1+2a_W^1\psi_A \\
-a_W^{*,1}\psi_0+a_P^{*,1}\psi_1-a_E^1\psi_2 &= b_1^*
\end{aligned}
$$

$$
\boxed{a_W^{*,1}=0} \quad \boxed{a_P^{*,1}=a_P^1+a_W^1} \quad \boxed{b_1^*=b_1+2a_W^1\psi_A}
$$

$$
\begin{aligned}
\frac{\psi_{Nx}+\psi_{Nx+1}}{2} &= \psi_B \\
\psi_{Nx+1} &= 2\psi_B-\psi_{Nx} \\
-a_W^{Nx}\psi_{Nx-1}+a_P^{Nx}\psi_{Nx}-a_E^{Nx}\psi_{Nx+1} &= b_{Nx} \\
-a_W^{Nx}\psi_{Nx-1}+a_P^{Nx}\psi_{Nx}-a_E^{Nx}(2\psi_B-\psi_{Nx}) &= b_{Nx} \\
-a_W^{Nx}\psi_{Nx-1}+(a_P^{Nx}+a_E^{Nx})\psi_{Nx} &= b_{Nx}+2a_E^{Nx}\psi_B \\
-a_W^{Nx}\psi_{Nx-1}+a_P^{*,Nx}\psi_{Nx}-a_E^{*,Nx}\psi_{Nx+1} &= b_{Nx}^*
\end{aligned}
$$

$$
\boxed{a_E^{*,Nx}=0} \quad \boxed{a_P^{*,Nx}=a_P^{Nx}+a_E^{Nx}} \quad \boxed{b_{Nx}^*=b_{Nx}+2a_E^{Nx}\psi_B}
$$

1.3.2. Neumman

$$
\begin{aligned}
\left.\frac{d\psi}{dx}\right|_w=\frac{\psi_1-\psi_0}{\Delta x_w} &= \mathrm{f}_A \\
\psi_0 &= \psi_1-\Delta x_w\mathrm{f}_A \\
-a_W^1\psi_0+a_P^1\psi_1-a_E^1\psi_2 &= b_1 \\
-a_W^1(\psi_1-\Delta x_w\mathrm{f}_A)+a_P^1\psi_1-a_E^1\psi_2 &= b_1 \\
(a_P^1-a_W^1)\psi_1-a_E^1\psi_2 &= b_1-a_W^1\Delta x_w\mathrm{f}_A \\
-a_W^{*,1}\psi_0+a_P^{*,1}\psi_1-a_E^1\psi_2 &= b_1^*
\end{aligned}
$$

$$
\boxed{a_W^{*,1}=0} \quad \boxed{a_P^{*,1}=a_P^1-a_W^1} \quad \boxed{b_1^*=b_1-a_W^1\Delta x_w\mathrm{f}_A}
$$

$$
\begin{aligned}
\left.\frac{d\psi}{dx}\right|_e=\frac{\psi_{Nx+1}-\psi_{Nx}}{\Delta x_e} &= \mathrm{f}_B \\
\psi_{Nx+1} &= \psi_{Nx}+\Delta x_e\mathrm{f}_B \\
-a_W^{Nx}\psi_{Nx-1}+a_P^{Nx}\psi_{Nx}-a_E^{Nx}\psi_{Nx+1} &= b_{Nx} \\
-a_W^{Nx}\psi_{Nx-1}+a_P^{Nx}\psi_{Nx}-a_E^{Nx}(\psi_{Nx}+\Delta x_e\mathrm{f}_B) &= b_{Nx} \\
-a_W^{Nx}\psi_{Nx-1}+(a_P^{Nx}-a_E^{Nx})\psi_{Nx} &= b_{Nx}+a_E^{Nx}\Delta x_e\mathrm{f}_B \\
-a_W^{Nx}\psi_{Nx-1}+a_P^{*,Nx}\psi_{Nx}-a_E^{*,Nx}\psi_{Nx+1} &= b_{Nx}^*
\end{aligned}
$$

$$
\boxed{a_E^{*,Nx}=0} \quad \boxed{a_P^{*,Nx}=a_P^{Nx}-a_E^{Nx}} \quad \boxed{b_{Nx}^*=b_{Nx}+a_E^{Nx}\Delta x_e\mathrm{f}_B}
$$

1.3.3. Ajuste en el almacenamiento CSR

$$
\left(\begin{array}{c|ccccc|c}
-a_W^{*,1} & a_P^{*,1} & -a_E^1 & 0 & 0 & 0 & 0 \\
0 & -a_W^2 & a_P^2 & -a_E^2 & 0 & 0 & 0 \\
0 & 0 & -a_W^3 & a_P^3 & -a_E^3 & 0 & 0 \\
0 & 0 & 0 & -a_W^4 & a_P^4 & -a_E^4 & 0 \\
0 & 0 & 0 & 0 & -a_W^5 & a_P^{*,5} & -a_E^{*,5}
\end{array}\right)
\begin{pmatrix}\psi_0\\ \psi_1\\ \psi_2\\ \psi_3\\ \psi_4\\ \psi_5\\ \psi_6\end{pmatrix}
=
\begin{pmatrix} b_1^* \\ b_2 \\ b_3 \\ b_4 \\ b_5^*\end{pmatrix}
$$

$$
AA = -a_W^{*,1} \quad a_P^{*,1} \quad -a_E^1 \quad -a_W^2 \quad a_P^2 \quad -a_E^2 \quad -a_W^3 \quad a_P^3 \quad -a_E^3 \quad -a_W^4 \quad a_P^4 \quad -a_E^4 \quad -a_W^5 \quad a_P^{*,5} \quad -a_E^{*,5}
$$

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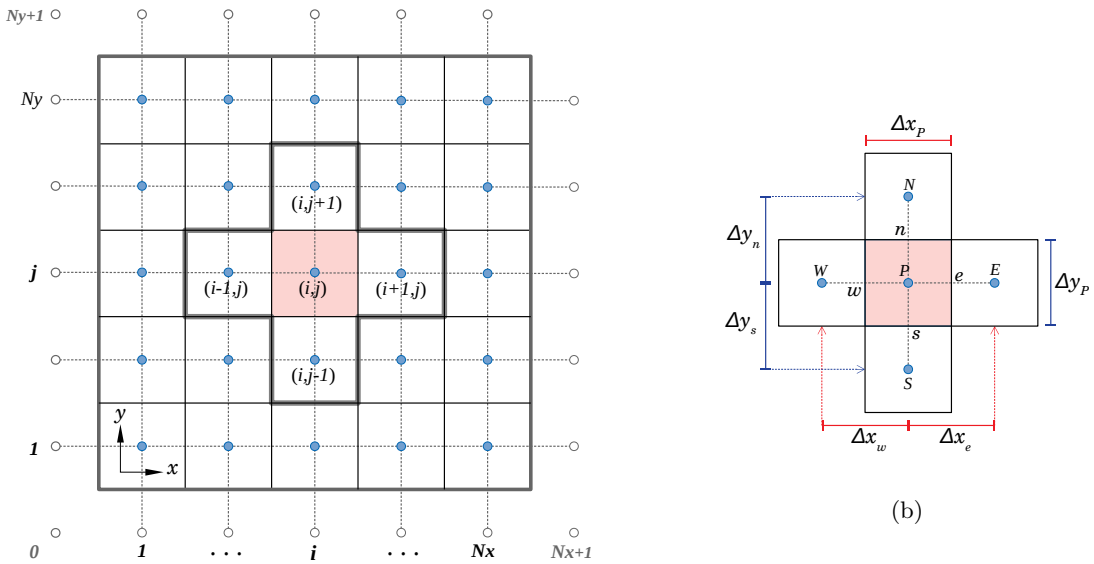
2. Modelo numérico 2D



(a) (b)

Figura 3: A figure with two subfigures

$$\boldsymbol{a_P}\psi_P = \boldsymbol{a_E}\psi_E + \boldsymbol{a_W}\psi_W + \boldsymbol{a_N}\psi_N + \boldsymbol{a_S}\psi_S + \boldsymbol{b_P} \quad \Longrightarrow \quad -\boldsymbol{a_S}\psi_S - \boldsymbol{a_W}\psi_W + \boldsymbol{a_P}\psi_P - \boldsymbol{a_E}\psi_E - \boldsymbol{a_N}\psi_N = \boldsymbol{b_P} \quad (1)$$

En este caso también usaremos un superíndice indicando el volumen al que corresponde la ecuación. En este caso dicho superíndice será $\boldsymbol{P \equiv i,j}$ como se muestra en la figura 3a. Entonces, para el volumen (i, j) se tiene:

$$-\boldsymbol{a_S^{i,j}}\psi_{i,j\text{-}1} - \boldsymbol{a_W^{i,j}}\psi_{i\text{-}1,j} + \boldsymbol{a_P^{i,j}}\psi_{i,j} - \boldsymbol{a_E^{i,j}}\psi_{i+1,j} - \boldsymbol{a_N^{i,j}}\psi_{i,j+1} = \boldsymbol{b_{i,j}}$$

2.1. Ecuaciones FVM

$$\begin{aligned}
\boldsymbol{S=(1,0), W=(0,1), P=(1,1), E=(2,1), N=(1,2)} &: -\boldsymbol{a_S^{1,1}}\psi_{1,0} - \boldsymbol{a_W^{1,1}}\psi_{0,1} + \boldsymbol{a_P^{1,1}}\psi_{1,1} - \boldsymbol{a_E^{1,1}}\psi_{2,1} - \boldsymbol{a_N^{1,1}}\psi_{1,2} = \boldsymbol{b_{1,1}} \\
\boldsymbol{S=(2,0), W=(1,1), P=(2,1), E=(3,1), N=(2,2)} &: -\boldsymbol{a_S^{2,1}}\psi_{2,0} - \boldsymbol{a_W^{2,1}}\psi_{1,1} + \boldsymbol{a_P^{2,1}}\psi_{2,1} - \boldsymbol{a_E^{2,1}}\psi_{3,1} - \boldsymbol{a_N^{2,1}}\psi_{2,2} = \boldsymbol{b_{2,1}} \\
\boldsymbol{S=(3,0), W=(2,1), P=(3,1), E=(4,1), N=(3,2)} &: -\boldsymbol{a_S^{3,1}}\psi_{3,0} - \boldsymbol{a_W^{3,1}}\psi_{2,1} + \boldsymbol{a_P^{3,1}}\psi_{3,1} - \boldsymbol{a_E^{3,1}}\psi_{4,1} - \boldsymbol{a_N^{3,1}}\psi_{3,2} = \boldsymbol{b_{3,1}} \\
\boldsymbol{S=(1,1), W=(0,2), P=(1,2), E=(2,2), N=(1,3)} &: -\boldsymbol{a_S^{1,2}}\psi_{1,1} - \boldsymbol{a_W^{1,2}}\psi_{0,2} + \boldsymbol{a_P^{1,2}}\psi_{1,2} - \boldsymbol{a_E^{1,2}}\psi_{2,2} - \boldsymbol{a_N^{1,2}}\psi_{1,3} = \boldsymbol{b_{1,2}} \\
\boldsymbol{S=(2,1), W=(1,2), P=(2,2), E=(3,2), N=(2,3)} &: -\boldsymbol{a_S^{2,2}}\psi_{2,1} - \boldsymbol{a_W^{2,2}}\psi_{1,2} + \boldsymbol{a_P^{2,2}}\psi_{2,2} - \boldsymbol{a_E^{2,2}}\psi_{3,2} - \boldsymbol{a_N^{2,2}}\psi_{2,3} = \boldsymbol{b_{2,2}} \\
\boldsymbol{S=(3,1), W=(2,2), P=(3,2), E=(4,2), N=(3,3)} &: -\boldsymbol{a_S^{3,2}}\psi_{3,1} - \boldsymbol{a_W^{3,2}}\psi_{2,2} + \boldsymbol{a_P^{3,2}}\psi_{3,2} - \boldsymbol{a_E^{3,2}}\psi_{4,2} - \boldsymbol{a_N^{3,2}}\psi_{3,3} = \boldsymbol{b_{3,2}} \\
\boldsymbol{S=(1,2), W=(0,3), P=(1,3), E=(2,3), N=(1,4)} &: -\boldsymbol{a_S^{1,3}}\psi_{1,2} - \boldsymbol{a_W^{1,3}}\psi_{0,3} + \boldsymbol{a_P^{1,3}}\psi_{1,3} - \boldsymbol{a_E^{1,3}}\psi_{2,3} - \boldsymbol{a_N^{1,3}}\psi_{1,4} = \boldsymbol{b_{1,3}} \\
\boldsymbol{S=(2,2), W=(1,3), P=(2,3), E=(3,3), N=(2,4)} &: -\boldsymbol{a_S^{2,3}}\psi_{2,2} - \boldsymbol{a_W^{2,3}}\psi_{1,3} + \boldsymbol{a_P^{2,3}}\psi_{2,3} - \boldsymbol{a_E^{2,3}}\psi_{3,3} - \boldsymbol{a_N^{2,3}}\psi_{2,4} = \boldsymbol{b_{2,3}} \\
\boldsymbol{S=(3,2), W=(2,3), P=(3,3), E=(4,3), N=(3,4)} &: -\boldsymbol{a_S^{3,3}}\psi_{3,2} - \boldsymbol{a_W^{3,3}}\psi_{2,3} + \boldsymbol{a_P^{3,3}}\psi_{3,3} - \boldsymbol{a_E^{3,3}}\psi_{4,3} - \boldsymbol{a_N^{3,3}}\psi_{3,4} = \boldsymbol{b_{3,3}}
\end{aligned}$$

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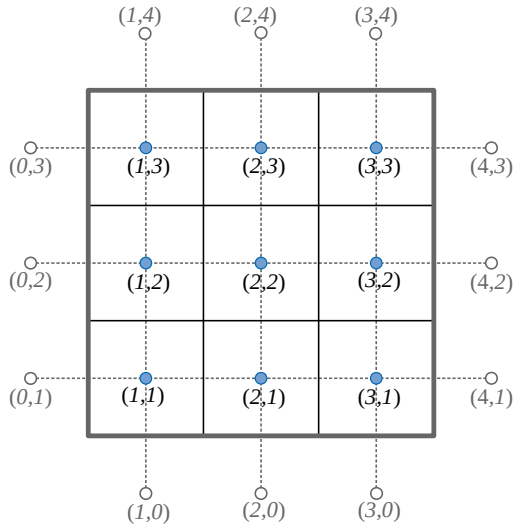

Figura 4: Malla en 2D.

$$
\left(\begin{array}{ccc|ccccccccc|ccc}
\boxed{-a_S^{1,1}} & 0 & \boxed{-a_W^{1,1}} & a_P^{1,1} & -a_E^{1,1} & 0 & -a_N^{1,1} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \boxed{-a_S^{2,1}} & 0 & -a_W^{2,1} & a_P^{2,1} & -a_E^{2,1} & 0 & -a_N^{2,1} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \boxed{-a_S^{3,1}} & 0 & -a_W^{3,1} & a_P^{3,1} & \boxed{-a_E^{3,1}} & 0 & -a_N^{3,1} & 0 & 0 & 0 & 0 & 0 \\
\hline
0 & 0 & 0 & -a_S^{1,2} & 0 & \boxed{-a_W^{1,2}} & a_P^{1,2} & -a_E^{1,2} & 0 & -a_N^{1,2} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -a_S^{2,2} & 0 & -a_W^{2,2} & a_P^{2,2} & -a_E^{2,2} & 0 & -a_N^{2,2} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -a_S^{3,2} & 0 & -a_W^{3,2} & -a_P^{3,2} & \boxed{-a_E^{3,2}} & 0 & -a_N^{3,2} & 0 & 0 \\
\hline
0 & 0 & 0 & 0 & 0 & 0 & -a_S^{1,3} & 0 & \boxed{-a_W^{1,3}} & -a_P^{1,3} & a_E^{1,3} & 0 & \boxed{-a_N^{1,3}} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -a_S^{2,3} & 0 & -a_W^{2,3} & a_P^{2,3} & -a_E^{2,3} & 0 & \boxed{-a_N^{2,3}} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -a_S^{3,3} & 0 & a_W^{3,3} & -a_P^{3,3} & \boxed{-a_E^{3,3}} & 0 & \boxed{-a_N^{3,3}}
\end{array}\right)
$$

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2.2. Almacenamiento de la matrix: CSR

Algoritmo 2 Formato CSR IJK 2D

```
INPUT: bi, ei, bj, ej
OUTPUT: AA, JA, IA
I := 0
irow := 0
IA[irow] := I
for i := bi to ei do
    for j := bj to ej do
        Calculate a_S^{i,j}, a_W^{i,j}, a_P^{i,j}, a_E^{i,j}, a_N^{i,j}
        AA[I] := a_S^{i,j}
        JA[I] := irow - NTx
        I := I + 1
        AA[I] := a_W^{i,j}
        JA[I] := irow - 1
        I := I + 1
        AA[I] := a_P^{i,j}
        JA[I] := irow
        I := I + 1
        AA[I] := a_E^{i,j}
        JA[I] := irow + 1
        I := I + 1
        AA[I] := a_N^{i,j}
        JA[I] := irow + NTx
        I := I + 1
        irow := irow + 1
        IA[irow] := I
    end for
end for
```

Algoritmo 3 Corrije: $\boldsymbol{a_W}$ y $\boldsymbol{a_E}$

```
OFFSET_X = Nd * NTx
for j := 0 to NTy-1 do do
    I := j * OFFSET_X
    JA[I+1] := 0
    I := I + OFFSET_X
    JA[I-2] := 0
end for
```

Algoritmo 4 Corrije: $\boldsymbol{a_S}$ y $\boldsymbol{a_N}$

```
OFFSET_Y := (NTx * (NTy - 1) + 1) * Nd
for i := 0 to NTx-1 do do
    I := i * Nd
    JA[I] := 0
    I := I + OFFSET_Y
    JA[I-1] := 0
end for
```

Matriz del sistema.

	0	1	2	3	4	5	6	7	8
0	$a_P^{1,1}$	$-a_E^{1,1}$	0	$-a_N^{1,1}$	0	0	0	0	0
1	$-a_W^{2,1}$	$a_P^{2,1}$	$-a_E^{2,1}$	0	$-a_N^{2,1}$	0	0	0	0
2	0	$-a_W^{3,1}$	$a_P^{3,1}$	0	0	$-a_N^{3,1}$	0	0	0
3	$-a_S^{1,2}$	0	0	$a_P^{1,2}$	$-a_E^{1,2}$	0	$-a_N^{1,2}$	0	0
4	0	$-a_S^{2,2}$	0	$-a_W^{2,2}$	$a_P^{2,2}$	$-a_E^{2,2}$	0	$-a_N^{2,2}$	0
5	0	0	$-a_S^{3,2}$	0	$a_W^{3,2}$	$-a_P^{3,2}$	0	0	$-a_N^{3,2}$
6	0	0	0	$-a_S^{1,3}$	0	0	$-a_P^{1,3}$	$a_E^{1,3}$	0
7	0	0	0	0	$-a_S^{2,3}$	0	$-a_W^{2,3}$	$a_P^{2,3}$	$-a_E^{2,3}$
8	0	0	0	0	0	$-a_S^{3,3}$	0	$a_W^{3,3}$	$-a_P^{3,3}$

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AA	$=$	$\boxed{-a_S^{1,1}}$	$\boxed{-a_W^{1,1}}$	$a_P^{1,1}$	$-a_E^{1,1}$	$-a_N^{1,1}$	$\boxed{-a_S^{2,1}}$	$-a_W^{2,1}$	$a_P^{2,1}$	$-a_E^{2,1}$	$-a_N^{2,1}$	$\boxed{-a_S^{3,1}}$	$-a_W^{3,1}$	$a_P^{3,1}$	$\boxed{-a_E^{3,1}}$	$-a_N^{3,1}$
		$-a_S^{1,1}$	$\boxed{-a_W^{1,1}}$	$a_P^{1,1}$	$-a_E^{1,1}$	$-a_N^{1,1}$	$-a_S^{2,1}$	$-a_W^{2,1}$	$a_P^{2,1}$	$-a_E^{2,1}$	$-a_N^{2,1}$	$-a_S^{3,1}$	$-a_W^{3,1}$	$a_P^{3,1}$	$\boxed{-a_E^{3,1}}$	$-a_N^{3,1}$
		$-a_S^{1,1}$	$\boxed{-a_W^{1,1}}$	$a_P^{1,1}$	$-a_E^{1,1}$	$\boxed{-a_N^{1,1}}$	$-a_S^{2,1}$	$-a_W^{2,1}$	$a_P^{2,1}$	$-a_E^{2,1}$	$\boxed{-a_N^{2,1}}$	$-a_S^{3,1}$	$-a_W^{3,1}$	$a_P^{3,1}$	$\boxed{-a_E^{3,1}}$	$\boxed{-a_N^{3,1}}$
JA	$=$	$\boxed{0}$	$\boxed{0}$	0	1	3	$\boxed{0}$	0	1	2	4	$\boxed{0}$	1	2	$\boxed{0}$	5
		0	$\boxed{0}$	3	4	6	1	3	4	5	7	2	4	5	$\boxed{0}$	8
		3	$\boxed{0}$	6	7	$\boxed{0}$	4	6	7	8	$\boxed{0}$	5	7	8	$\boxed{0}$	$\boxed{0}$
IA	$=$	0	5	10	15	20	25	30	35	40	45					

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2.3. Condiciones de frontera

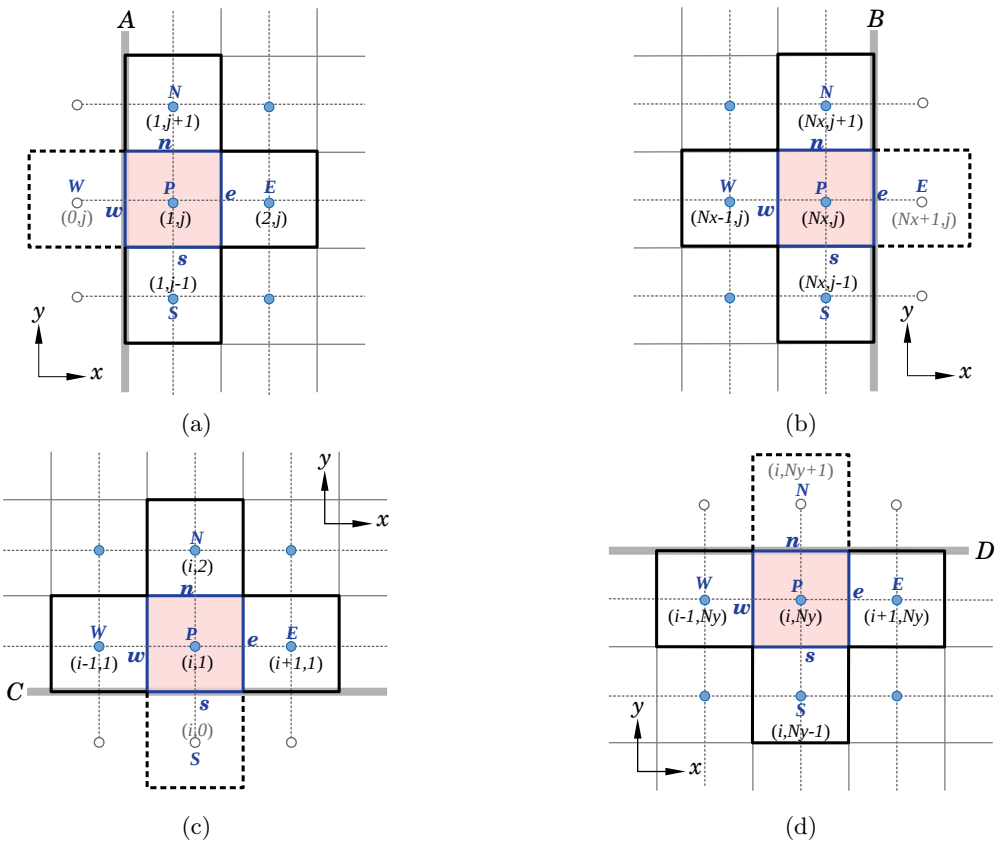

Figura 5: Condiciones de frontera

2.3.1. Dirichlet

$$
\begin{aligned}
\frac{\psi_{0,j}+\psi_{1,j}}{2} &= \psi_A \\
\psi_{0,j} &= 2\psi_A - \psi_{1,j} \\
-a_S^{1,j}\psi_{1,j-1} - a_W^{1,j}\psi_{0,j} + a_P^{1,j}\psi_{1,j} - a_E^{1,j}\psi_{2,j} - a_N^{1,j}\psi_{1,j+1} &= b_{1,j} \\
-a_S^{1,j}\psi_{1,j-1} - a_W^{1,j}(2\psi_A - \psi_{1,j}) + a_P^{1,j}\psi_{1,j} - a_E^{1,j}\psi_{2,j} - a_N^{1,j}\psi_{1,j+1} &= b_{1,j} \\
-a_S^{1,j}\psi_{1,j-1} + (a_P^{1,j} + a_W^{1,j})\psi_{1,j} - a_E^{1,j}\psi_{2,j} - a_N^{1,j}\psi_{1,j+1} &= b_{1,j} + 2a_W^{1,j}\psi_A \\
-a_S^{1,j}\psi_{1,j-1} - a_W^{*,1,j}\psi_{0,j} + a_P^{*,1,j}\psi_{1,j} - a_E^{1,j}\psi_{2,j} - a_N^{1,j}\psi_{1,j+1} &= b_{1,j}^*
\end{aligned}
$$

$$\boxed{a_W^{*,1,j} = 0} \qquad \boxed{a_P^{*,1,j} = a_P^{1,j} + a_W^{1,j}} \qquad \boxed{b_{1,j}^* = b_{1,j} + 2a_W^{1,j}\psi_A}$$

$$
\begin{aligned}
\frac{\psi_{Nx,j}+\psi_{Nx+1,j}}{2} &= \psi_B \\
\psi_{Nx+1,j} &= 2\psi_B-\psi_{Nx,j} \\
-\boldsymbol{a}_S^{Nx,j}\psi_{Nx,j-1}-\boldsymbol{a}_W^{Nx,j}\psi_{Nx-1,j}+\boldsymbol{a}_P^{Nx,j}\psi_{Nx,j}-\boldsymbol{a}_E^{Nx,j}\psi_{Nx+1,j}-\boldsymbol{a}_N^{Nx,j}\psi_{Nx,j+1} &= \boldsymbol{b}_{Nx,j} \\
-\boldsymbol{a}_S^{Nx,j}\psi_{Nx,j-1}-\boldsymbol{a}_W^{Nx,j}\psi_{Nx-1,j-1}+\boldsymbol{a}_P^{Nx,j}\psi_{Nx,j}-\boldsymbol{a}_E^{Nx,j}(2\psi_B-\psi_{Nx,j})-\boldsymbol{a}_N^{Nx,j}\psi_{Nx,j+1} &= \boldsymbol{b}_{Nx,j} \\
-\boldsymbol{a}_S^{Nx,j}\psi_{Nx,j-1}-\boldsymbol{a}_W^{Nx,j}\psi_{Nx-1,j}+(\boldsymbol{a}_P^{Nx,j}+\boldsymbol{a}_E^{Nx,j})\psi_{Nx,j}-\boldsymbol{a}_N^{Nx,j}\psi_{Nx,j+1} &= \boldsymbol{b}_{Nx,j}+2\boldsymbol{a}_E^{Nx,j}\psi_B \\
-\boldsymbol{a}_S^{Nx,j}\psi_{Nx,j-1}-\boldsymbol{a}_W^{Nx,j}\psi_{Nx-1,j}+\boldsymbol{a}_P^{*,Nx,j}\psi_{Nx,j}-\boldsymbol{a}_E^{*,Nx,j}\psi_{Nx+1,j}-\boldsymbol{a}_N^{Nx,j}\psi_{Nx,j+1} &= \boldsymbol{b}_{Nx,j}^*
\end{aligned}
$$

$$\boxed{\boldsymbol{a}_E^{*,Nx,j}=0} \qquad \boxed{\boldsymbol{a}_P^{*,Nx,j}=\boldsymbol{a}_P^{Nx,j}+\boldsymbol{a}_E^{Nx,j}} \qquad \boxed{\boldsymbol{b}_{Nx,j}^*=\boldsymbol{b}_{Nx,j}+2\boldsymbol{a}_E^{Nx,j}\psi_B}$$

$$
\begin{aligned}
\frac{\psi_{i,0}+\psi_{i,1}}{2} &= \psi_C \\
\psi_{i,0} &= 2\psi_C-\psi_{i,1} \\
-\boldsymbol{a}_S^{i,1}\psi_{i,0}-\boldsymbol{a}_W^{i,1}\psi_{i-1,1}+\boldsymbol{a}_P^{i,1}\psi_{i,1}-\boldsymbol{a}_E^{i,1}\psi_{i+1,1}-\boldsymbol{a}_N^{i,1}\psi_{i,2} &= \boldsymbol{b}_{i,1} \\
-\boldsymbol{a}_S^{i,1}(2\psi_C-\psi_{i,1})-\boldsymbol{a}_W^{i,1}\psi_{i-1,1}+\boldsymbol{a}_P^{i,1}\psi_{i,1}-\boldsymbol{a}_E^{i,1}\psi_{i+1,1}-\boldsymbol{a}_N^{i,1}\psi_{i,2} &= \boldsymbol{b}_{i,1} \\
-\boldsymbol{a}_W^{i,1}\psi_{i-1,1}+(\boldsymbol{a}_P^{i,1}+\boldsymbol{a}_S^{i,1})\psi_{i,1}-\boldsymbol{a}_E^{i,1}\psi_{i+1,1}-\boldsymbol{a}_N^{i,1}\psi_{i,2} &= \boldsymbol{b}_{i,1}+2\boldsymbol{a}_S^{i,1}\psi_C \\
-\boldsymbol{a}_S^{*,i,1}\psi_{i,0}-\boldsymbol{a}_W^{i,1}\psi_{i-1,1}+\boldsymbol{a}_P^{*,i,1}\psi_{i,1}-\boldsymbol{a}_E^{i,1}\psi_{i+1,1}-\boldsymbol{a}_N^{i,1}\psi_{i,2} &= \boldsymbol{b}_{i,1}^*
\end{aligned}
$$

$$\boxed{\boldsymbol{a}_S^{*,i,1}=0} \qquad \boxed{\boldsymbol{a}_P^{*,i,1}=\boldsymbol{a}_P^{i,1}+\boldsymbol{a}_S^{i,1}} \qquad \boxed{\boldsymbol{b}_{i,1}^*=\boldsymbol{b}_{i,1}+2\boldsymbol{a}_S^{i,1}\psi_C}$$

$$
\begin{aligned}
\frac{\psi_{i,Ny}+\psi_{i,Ny+1}}{2} &= \psi_D \\
\psi_{i,Ny+1} &= 2\psi_D-\psi_{i,Ny} \\
-\boldsymbol{a}_S^{i,Ny}\psi_{i,Ny-1}-\boldsymbol{a}_W^{i,Ny}\psi_{i-1,Ny}+\boldsymbol{a}_P^{i,Ny}\psi_{i,Ny}-\boldsymbol{a}_E^{i,Ny}\psi_{i+1,Ny}-\boldsymbol{a}_N^{i,Ny}\psi_{i,Ny+1} &= \boldsymbol{b}_{i,Ny} \\
-\boldsymbol{a}_S^{i,Ny}\psi_{i,Ny-1}-\boldsymbol{a}_W^{i,Ny}\psi_{i-1,Ny}+\boldsymbol{a}_P^{i,Ny}\psi_{i,Ny}-\boldsymbol{a}_E^{i,Ny}\psi_{i+1,Ny}-\boldsymbol{a}_N^{i,Ny}(2\psi_D-\psi_{i,Ny}) &= \boldsymbol{b}_{i,Ny} \\
-\boldsymbol{a}_S^{i,Ny}\psi_{i,Ny-1}-\boldsymbol{a}_W^{i,Ny}\psi_{i-1,Ny}+(\boldsymbol{a}_P^{i,Ny}+\boldsymbol{a}_N^{i,Ny})\psi_{i,Ny}-\boldsymbol{a}_E^{i,Ny}\psi_{i+1,Ny} &= \boldsymbol{b}_{i,Ny}+2\boldsymbol{a}_N^{i,Ny}\psi_D \\
-\boldsymbol{a}_S^{i,Ny}\psi_{i,Ny-1}-\boldsymbol{a}_W^{i,Ny}\psi_{i-1,Ny}+\boldsymbol{a}_P^{*,i,Ny}\psi_{i,Ny}-\boldsymbol{a}_E^{i,Ny}\psi_{i+1,Ny}-\boldsymbol{a}_N^{*,i,Ny}\psi_{i,Ny+1} &= \boldsymbol{b}_{i,Ny}^*
\end{aligned}
$$

$$\boxed{\boldsymbol{a}_N^{*,i,Ny}=0} \qquad \boxed{\boldsymbol{a}_P^{*,i,Ny}=\boldsymbol{a}_P^{i,Ny}+\boldsymbol{a}_N^{i,Ny}} \qquad \boxed{\boldsymbol{b}_{i,Ny}^*=\boldsymbol{b}_{i,Ny}+2\boldsymbol{a}_N^{i,Ny}\psi_D}$$

2.3.2. Neumman

$$
\begin{aligned}
\left.\frac{\partial\psi}{\partial x}\right|_w=\frac{\psi_{1,j}-\psi_{0,j}}{\Delta x_w} &= \mathrm{f}_A \\
\psi_{0,j} &= \psi_{1,j}-\Delta x_w\mathrm{f}_A \\
-\boldsymbol{a}_S^{1,j}\psi_{1,j-1}-\boldsymbol{a}_W^{1,j}\psi_{0,j}+\boldsymbol{a}_P^{1,j}\psi_{1,j}-\boldsymbol{a}_E^{1,j}\psi_{2,j}-\boldsymbol{a}_N^{1,j}\psi_{1,j+1} &= \boldsymbol{b}_{1,j} \\
-\boldsymbol{a}_S^{1,j}\psi_{1,j-1}-\boldsymbol{a}_W^{1,j}(\psi_{1,j}-\Delta x_w\mathrm{f}_A)+\boldsymbol{a}_P^{1,j}\psi_{1,j}-\boldsymbol{a}_E^{1,j}\psi_{2,j}-\boldsymbol{a}_N^{1,j}\psi_{1,j+1} &= \boldsymbol{b}_{1,j} \\
-\boldsymbol{a}_S^{1,j}\psi_{1,j-1}+(\boldsymbol{a}_P^{1,j}-\boldsymbol{a}_W^{1,j})\psi_{1,j}-\boldsymbol{a}_E^{1,j}\psi_{2,j}-\boldsymbol{a}_N^{1,j}\psi_{1,j+1} &= \boldsymbol{b}_{1,j}-\boldsymbol{a}_W^{1,j}\Delta x_w\mathrm{f}_A \\
-\boldsymbol{a}_S^{1,j}\psi_{1,j-1}-\boldsymbol{a}_W^{*,1,j}\psi_{0,j}+\boldsymbol{a}_P^{*,1,j}\psi_{1,j}-\boldsymbol{a}_E^{1,j}\psi_{2,j}-\boldsymbol{a}_N^{1,j}\psi_{1,j+1} &= \boldsymbol{b}_{1,j}^*
\end{aligned}
$$

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$$\boxed{a_W^{*,1,j} = 0} \qquad \boxed{a_P^{*,1,j} = a_P^{1,j} - a_W^{1,j}} \qquad \boxed{b_{1,j}^{*} = b_{1,j} - a_W^{1,j}\Delta x_w \mathrm{f}_A}$$

$$\begin{aligned}
\left.\frac{\partial \psi}{\partial x}\right|_e = \frac{\psi_{Nx+1,j} - \psi_{Nx,j}}{\Delta x_e} &= \mathrm{f}_B \\
\psi_{Nx+1,j} &= \psi_{Nx,j} + \Delta x_e \mathrm{f}_B \\
-a_S^{Nx,j}\psi_{Nx,j-1} - a_W^{Nx,j}\psi_{Nx-1,j} + a_P^{Nx,j}\psi_{Nx,j} - a_E^{Nx,j}\psi_{Nx+1,j} - a_N^{Nx,j}\psi_{Nx,j+1} &= b_{Nx,j} \\
-a_S^{Nx,j}\psi_{Nx,j-1} - a_W^{Nx,j}\psi_{Nx-1,j-1} + a_P^{Nx,j}\psi_{Nx,j} - a_E^{Nx,j}(\psi_{Nx,j} + \Delta x_e \mathrm{f}_B) - a_N^{Nx,j}\psi_{Nx,j+1} &= b_{Nx,j} \\
-a_S^{Nx,j}\psi_{Nx,j-1} - a_W^{Nx,j}\psi_{Nx-1,j} + (a_P^{Nx,j} - a_E^{Nx,j})\psi_{Nx,j} - a_N^{Nx,j}\psi_{Nx,j+1} &= b_{Nx,j} + a_E^{Nx,j}\Delta x_e \mathrm{f}_B \\
-a_S^{Nx,j}\psi_{Nx,j-1} - a_W^{Nx,j}\psi_{Nx-1,j} + a_P^{*,Nx,j}\psi_{Nx,j} - a_E^{*,Nx,j}\psi_{Nx+1,j} - a_N^{Nx,j}\psi_{Nx,j+1} &= b_{Nx,j}^{*}
\end{aligned}$$

$$\boxed{a_E^{*,Nx,j} = 0} \qquad \boxed{a_P^{*,Nx,j} = a_P^{Nx,j} - a_E^{Nx,j}} \qquad \boxed{b_{Nx,j}^{*} = b_{Nx,j} + a_E^{Nx,j}\Delta x_e \mathrm{f}_B}$$

$$\begin{aligned}
\left.\frac{\partial \psi}{\partial y}\right|_s = \frac{\psi_{i,1} - \psi_{i,0}}{\Delta y_s} &= \mathrm{f}_C \\
\psi_{i,0} &= \psi_{i,1} - \Delta y_s \mathrm{f}_C \\
-a_S^{i,1}\psi_{i,0} - a_W^{i,1}\psi_{i-1,1} + a_P^{i,1}\psi_{i,1} - a_E^{i,1}\psi_{i+1,1} - a_N^{i,1}\psi_{i,2} &= b_{i,1} \\
-a_S^{i,1}(\psi_{i,1} - \Delta y_s \mathrm{f}_C) - a_W^{i,1}\psi_{i-1,1} + a_P^{i,1}\psi_{i,1} - a_E^{i,1}\psi_{i+1,1} - a_N^{i,1}\psi_{i,2} &= b_{i,1} \\
-a_W^{i,1}\psi_{i-1,1} + (a_P^{i,1} - a_S^{i,1})\psi_{i,1} - a_E^{i,1}\psi_{i+1,1} - a_N^{i,1}\psi_{i,2} &= b_{i,1} - a_S^{i,1}\Delta y_s \mathrm{f}_C \\
-a_S^{*,i,1}\psi_{i,0} - a_W^{i,1}\psi_{i-1,1} + a_P^{*,i,1}\psi_{i,1} - a_E^{i,1}\psi_{i+1,1} - a_N^{i,1}\psi_{i,2} &= b_{i,1}^{*}
\end{aligned}$$

$$\boxed{a_S^{*,i,1} = 0} \qquad \boxed{a_P^{*,i,1} = a_P^{i,1} - a_S^{i,1}} \qquad \boxed{b_{i,1}^{*} = b_{i,1} - a_S^{i,1}\Delta y_s \mathrm{f}_C}$$

$$\begin{aligned}
\left.\frac{\partial \psi}{\partial y}\right|_n = \frac{\psi_{i,Ny+1} - \psi_{i,Ny}}{\Delta y_n} &= \mathrm{f}_D \\
\psi_{i,Ny+1} &= \psi_{i,Ny} + \Delta y_n \mathrm{f}_D \\
-a_S^{i,Ny}\psi_{i,Ny-1} - a_W^{i,Ny}\psi_{i-1,Ny} + a_P^{i,Ny}\psi_{i,Ny} - a_E^{i,Ny}\psi_{i+1,Ny} - a_N^{i,Ny}\psi_{i,Ny+1} &= b_{i,Ny} \\
-a_S^{i,Ny}\psi_{i,Ny-1} - a_W^{i,Ny}\psi_{i-1,Ny} + a_P^{i,Ny}\psi_{i,Ny} - a_E^{i,Ny}\psi_{i+1,Ny} - a_N^{i,Ny}(\psi_{i,Ny} + \Delta y_n \mathrm{f}_D) &= b_{i,Ny} \\
-a_S^{i,Ny}\psi_{i,Ny-1} - a_W^{i,Ny}\psi_{i-1,Ny} + (a_P^{i,Ny} - a_N^{i,Ny})\psi_{i,Ny} - a_E^{i,Ny}\psi_{i+1,Ny} &= b_{i,Ny} + a_N^{i,Ny}\Delta y_n \mathrm{f}_D \\
-a_S^{i,Ny}\psi_{i,Ny-1} - a_W^{i,Ny}\psi_{i-1,Ny} + a_P^{*,i,Ny}\psi_{i,Ny} - a_E^{i,Ny}\psi_{i+1,Ny} - a_N^{*,i,Ny}\psi_{i,Ny+1} &= b_{i,Ny}^{*}
\end{aligned}$$

$$\boxed{a_N^{*,i,Ny} = 0} \qquad \boxed{a_P^{*,i,Ny} = a_P^{i,Ny} - a_N^{i,Ny}} \qquad \boxed{b_{i,Ny}^{*} = b_{i,Ny} + a_N^{i,Ny}\Delta y_n \mathrm{f}_D}$$

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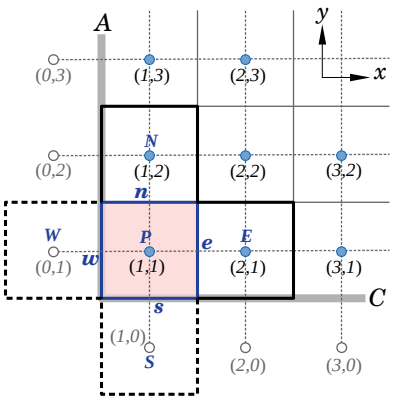

Figura 6: doble condición de frontera.

2.3.3. Doble condición de frontera

$$
\begin{aligned}
\frac{\psi_{0,1}+\psi_{1,1}}{2} &= \psi_A \\
\psi_{0,1} &= 2\psi_A - \psi_{1,1} \\
\left.\frac{\partial\psi}{\partial y}\right|_s = \frac{\psi_{1,1}-\psi_{1,0}}{\Delta y_s} &= \mathrm{f}_C \\
\psi_{1,0} &= \psi_{1,1} - \Delta y_s \mathrm{f}_C \\
-\boldsymbol{a}_S^{1,1}\psi_{1,0} - \boldsymbol{a}_W^{1,1}\psi_{0,1} + \boldsymbol{a}_P^{1,1}\psi_{1,1} - \boldsymbol{a}_E^{1,1}\psi_{2,1} - \boldsymbol{a}_N^{1,1}\psi_{1,2} &= \boldsymbol{b}_{1,1} \\
-\boldsymbol{a}_S^{1,1}(\psi_{1,1} - \Delta y_s \mathrm{f}_C) - \boldsymbol{a}_W^{1,1}(2\psi_A - \psi_{1,1}) + \boldsymbol{a}_P^{1,1}\psi_{1,1} - \boldsymbol{a}_E^{1,1}\psi_{2,1} - \boldsymbol{a}_N^{1,1}\psi_{1,2} &= \boldsymbol{b}_{1,1} \\
(\boldsymbol{a}_P^{1,1} + \boldsymbol{a}_W^{1,1} - \boldsymbol{a}_S^{1,1})\psi_{1,1} - \boldsymbol{a}_E^{1,1}\psi_{2,1} - \boldsymbol{a}_N^{1,1}\psi_{1,2} &= \boldsymbol{b}_{1,1} + 2\boldsymbol{a}_W^{1,1}\psi_A - \boldsymbol{a}_S^{1,1}\Delta y_s \mathrm{f}_C \\
-\boldsymbol{a}_S^{*,1,1}\psi_{1,0} - \boldsymbol{a}_W^{*,1,1}\psi_{0,1} + \boldsymbol{a}_P^{*,1,1}\psi_{1,1} - \boldsymbol{a}_E^{1,1}\psi_{2,1} - \boldsymbol{a}_N^{1,1}\psi_{1,2} &= \boldsymbol{b}_{1,1}^*
\end{aligned}
$$

$$
\boxed{\boldsymbol{a}_S^{*,1,1} = 0} \qquad \boxed{\boldsymbol{a}_W^{*,1,1} = 0} \qquad \boxed{\boldsymbol{a}_P^{*,1,1} = \boldsymbol{a}_P^{1,1} + \boldsymbol{a}_W^{1,1} - \boldsymbol{a}_S^{1,1}} \qquad \boxed{\boldsymbol{b}_{1,1}^* = \boldsymbol{b}_{1,1} + 2\boldsymbol{a}_W^{1,1}\psi_A - \boldsymbol{a}_S^{1,1}\Delta y_s \mathrm{f}_C}
$$

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2.3.4. Ajuste en el almacenamiento CSR

$$
\left(\begin{array}{ccc|ccccccccc|ccc}
-a_S^{*,1,1} & 0 & -a_W^{*,1,1} & a_P^{*,1,1} & -a_E^{1,1} & 0 & -a_N^{1,1} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -a_S^{*,2,1} & 0 & -a_W^{2,1} & a_P^{*,2,1} & -a_E^{2,1} & 0 & -a_N^{2,1} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -a_S^{*,3,1} & 0 & -a_W^{3,1} & a_P^{*,3,1} & -a_E^{*,3,1} & 0 & -a_N^{3,1} & 0 & 0 & 0 & 0 & 0 & 0 \\
\hline
0 & 0 & 0 & -a_S^{1,2} & 0 & -a_W^{*,1,2} & a_P^{*,1,2} & -a_E^{1,2} & 0 & -a_N^{1,2} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -a_S^{2,2} & 0 & -a_W^{2,2} & a_P^{2,2} & -a_E^{2,2} & 0 & -a_N^{2,2} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -a_S^{3,2} & 0 & -a_W^{3,2} & -a_P^{*,3,2} & -a_E^{*,3,2} & 0 & -a_N^{3,2} & 0 & 0 & 0 \\
\hline
0 & 0 & 0 & 0 & 0 & 0 & -a_S^{1,3} & 0 & -a_W^{*,1,3} & -a_P^{*,1,3} & a_E^{1,3} & 0 & -a_N^{*,1,3} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -a_S^{2,3} & 0 & -a_W^{2,3} & a_P^{*,2,3} & -a_E^{2,3} & 0 & -a_N^{*,2,3} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -a_S^{3,3} & 0 & a_W^{3,3} & -a_P^{*,3,3} & -a_E^{*,3,3} & 0 & -a_N^{*,3,3}
\end{array}\right)
$$

$$
\begin{array}{rcccccccccccccccc}
AA & = & -a_S^{*,1,1} & -a_W^{*,1,1} & a_P^{*,1,1} & -a_E^{1,1} & -a_N^{1,1} & -a_S^{*,2,1} & -a_W^{2,1} & a_P^{*,2,1} & -a_E^{2,1} & -a_N^{2,1} & -a_S^{*,3,1} & -a_W^{3,1} & a_P^{*,3,1} & -a_E^{*,3,1} & -a_N^{3,1} \\
 & & -a_S^{1,1} & -a_W^{*,1,1} & a_P^{*,1,1} & -a_E^{1,1} & -a_N^{1,1} & -a_S^{2,1} & -a_W^{2,1} & a_P^{2,1} & -a_E^{2,1} & -a_N^{2,1} & -a_S^{3,1} & -a_W^{3,1} & a_P^{*,3,1} & -a_E^{*,3,1} & -a_N^{3,1} \\
 & & -a_S^{1,1} & -a_W^{*,1,1} & a_P^{*,1,1} & -a_E^{1,1} & -a_N^{*,1,1} & -a_S^{2,1} & -a_W^{2,1} & a_P^{*,2,1} & -a_E^{2,1} & -a_N^{*,2,1} & -a_S^{3,1} & -a_W^{3,1} & a_P^{*,3,1} & -a_E^{*,3,1} & -a_N^{*,3,1}
\end{array}
$$

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3. Numerical Schemes

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